Year 13 Maths - Moments

Similar questions - Ladders and rods against walls and strings

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A uniform ladder AB of length 6 m and mass 20 kg stands with A on rough horizontal ground and B against a smooth vertical wall. The ladder makes an angle of 65° with the ground. The coefficient of friction between the ladder and the ground is 0.4. A man of mass 80 kg stands on the ladder at a distance d m from A (measured along the ladder). Take $g = 9.8$.

(a) Draw a diagram showing all the forces acting on the ladder. **[2]**

(b) Given that the ladder is in limiting equilibrium, find d. **[6]**

(c) Suggest one improvement to the model. **[1]**

2 A uniform rod AB of mass 4 kg and length 3 m rests in a vertical plane with A on rough horizontal ground. The rod is inclined at 60° to the horizontal. A particle of mass 1 kg is attached at B. The rod is held in limiting equilibrium by a light inextensible horizontal string attached to the point C on the rod, where $AC = x$ m, the other end being fixed to a point vertically above A. The coefficient of friction between the rod and the ground is 0.6. Take $g = 9.8$.

(a) Show that the tension in the string is $\dfrac{9g}{\sqrt{3}x}$ N. **[4]**

(b) Find x. **[4]**

Question 1

(a) At A: normal reaction N upwards and friction F towards the wall. At B: horizontal reaction S from the smooth wall. Weights $20g$ at the midpoint and $80g$ at distance d

(b) Vertically $N = 100g$. Limiting friction $F = 0.4 \times 100g = 40g$, and horizontally $S = F = 40g$. Moments about A: $S \times 6\sin 65^\circ = 20g \times 3\cos 65^\circ + 80g \times d\cos 65^\circ$. $40g \times 6 \times 0.9063 = g(60 \times 0.4226 + 80 \times 0.4226d)$, so $217.5 = 25.36 + 33.81d$ and $d = 5.68$ m

(c) Include the thickness of the ladder / treat the man as a rigid body, or allow for friction at the wall

Question 2

(a) Moments about A: tension (horizontal) acts at height $x\sin 60^\circ$ above A: $Tx\sin 60^\circ = 4g \times 1.5\cos 60^\circ + 1g \times 3\cos 60^\circ = 3g + 1.5g = 4.5g$. So $T = \dfrac{4.5g}{x\sin 60^\circ} = \dfrac{9g}{\sqrt{3}x}$

(b) Vertical: $N = 5g$. Limiting: $F = 0.6 \times 5g = 3g$ and horizontally $F = T$. So $\dfrac{9g}{\sqrt{3}x} = 3g$ giving $x = \sqrt{3} = 1.73$ m